PIC 10A 1A

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Today…

- Discussion on HW4
- Arrays
  - HW3 Solution (Array Version)

Discussion on HW4

HW4

- Exercise 2:
  - I shared the solution to the Num2Rome solution before
  - You can get some hints from “Strategy A”

```
string romans = "IVXLCDM", output = "";
int place = 0;
while (n > 0) {
    if (n % 10 > 0){
        if (n % 10 == 9) { // "IX", "XC", or "CM" (02, 24, 46)
            output = romans[place*2] + (romans[place*2 + 2] + output);
            n -= 9;
        }
        else if (n % 10 == 4) { // "IV", "XL", "CD" (01, 23, 45)
            output = romans[place * 2] + (romans[place * 2 + 1] + output);
            n -= 4;
        }
        while (n % 5 > 0) { // #: 1,2,3, 6,7,8 (append I, X, C or M)
            output = romans[place * 2] + output;
            --n;
        }
        if (n % 10 == 5) { // V, L, or D
            output = romans[place * 2 + 1] + output;
            n -= 5;
        }
    }
    n /= 10;
    ++place;
}
cout << output << endl;
```

←Replaced by a function call

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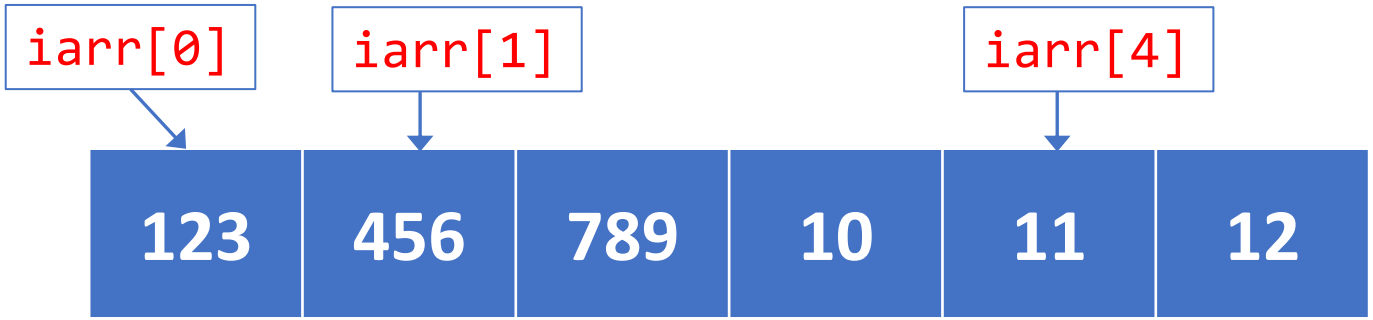
Arrays

Arrays

- An int array of size 6 may look like (in the memory):



- "iarr" is the name of the array object
- This "iarr" actually "points to" the address of the part of the memory that stores the value 123 (or, iarr[0])
- Accessing the k-th element of iarr → iarr[k] (subscript operator)

HW3 Solution (Array Version)

- Here's a HW3 Solution using an array
- In fact, this will help you do HW5
- (code review)

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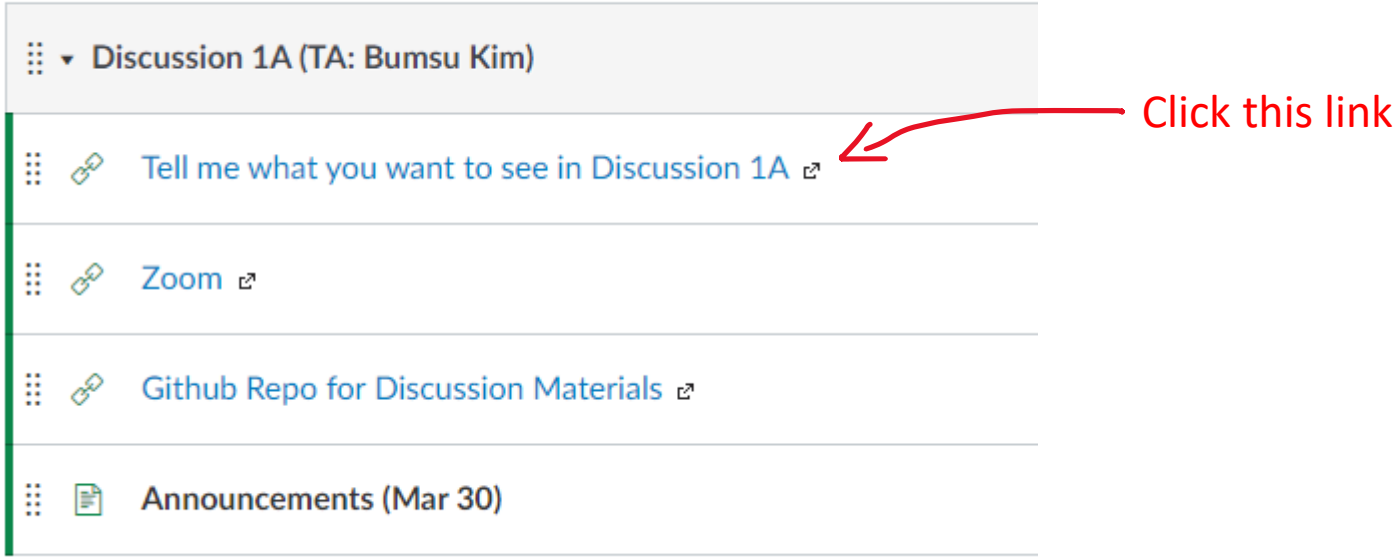
Your Feedback is welcome

- Don’t hesitate to give a feedback on the discussion
- You can use a link in BruinLearn (Google Form)